

Prescribing Information

For the use of Registered Medical Practitioner  
or Hospital or Laboratory Only.

**BUPISPINE HEAVY**  
0.5% w/v Solution for Injection

Bupivacaine Hydrochloride in Dextrose Injection USP

**COMPOSITION :**

Each ml contains :  
Bupivacaine Hydrochloride USP  
equivalent to Anhydrous Bupivacaine Hydrochloride 5.0 mg.  
Dextrose USP 80.0 mg. (Monohydrate)  
Water for Injections BP q.s.  
Other excipients, see EXCIPIENTS

**DESCRIPTION :**

BUPISPINE HEAVY 0.5% w/v solution for Injection-Bupivacaine hydrochloride in dextrose injection USP is clear & colourless solution.  
Each mL contains 5.280 mg of Bupivacaine HCl (Monohydrate form) which is equivalent to 5.000 mg of anhydrous Bupivacaine HCl.

**CLINICAL PHARMACOLOGY :**

**Pharmacodynamic properties :**

Pharmacotherapeutic group (ATC code) : N01B B01  
Bupivacaine is a long acting local anaesthetic agent of the amide type.  
Moderate muscular relaxation of lower extremities.  
Motor blockade of the abdominal muscles.

BUPISPINE HEAVY is hyperbaric and its initial spread in the intrathecal space is affected by gravity.

**Pharmacokinetic properties :**

Rapid onset of action and long duration i.e. T<sub>1/2</sub> - T<sub>1/2</sub> segments - duration 2 - 3 hours.  
Muscular relaxation of lower extremities lasts 2 - 2.5 hours.  
Blockade of the abdominal muscles lasts 45 - 60 minutes. The duration of motor blockade does not exceed duration of analgesia.  
In children the pharmacokinetics are similar to that in adults.

**INDICATIONS :**

Spinal anaesthesia for surgery, e.g. urological surgery and surgery on the lower limbs lasting 2 - 3 hours, abdominal surgery lasting 45-60 minutes.

**DOSAGE AND ADMINISTRATION :**

**Dosage :**  
BUPISPINE HEAVY should only be used by clinicians with experience of regional anaesthesia or under their supervision. The lowest possible dose for adequate anaesthesia should be used. The doses given below are guides for adults and the dosage should be adjusted to the individual patient.  
The dose should be reduced in the elderly and in patients in the late stages of pregnancy.

| Indication                                      | Dose (ml)  | Dose (mg)   | Time to onset of effect in minutes (approx.) | Duration of effect in hours (approx.) |
|-------------------------------------------------|------------|-------------|----------------------------------------------|---------------------------------------|
| Urological surgery                              | 1.5 - 3 ml | 7.5 - 15 mg | 5 - 8 min                                    | 2 - 3 hours                           |
| Surgery on lower limbs, including hip surgery   | 2 - 4 ml   | 10 - 20 mg  | 5 - 8 min                                    | 2 - 3 hours                           |
| Abdominal surgery (including caesarean section) | 2 - 4 ml   | 10 - 20 mg  | 5 - 8 min                                    | 45 - 60 min                           |

The recommended site of injection is below L3 - L4 intervertebral space.

There is currently no experience of doses higher than 20 mg.  
A spinal injection is given only after the subarachnoid space has been clearly identified by means of lumbar puncture (clear cerebrospinal fluid runs out via the spinal needle or is seen on aspiration). In the event of unsuccessful anaesthesia, a new attempt to administer the drug should only be made by injecting at a different level and with a smaller volume. One cause of lack of effect may be poor intrathecal distribution of the drug, and this can be helped by altering the patient's position.

**Method of administration :**

**Route of administration :** For intrathecal injection

**CONTRAINDICATIONS :**

- Hypersensitivity to the active substance, to any of the excipients, to local anaesthetics of the amide type.  
Intrathecal anaesthesia, regardless of the local anaesthetic used, has its own contraindications, which include :
- Active disease of the central nervous system such as meningitis, poliomyelitis, intracranial haemorrhage, sub-acute combined degeneration of the cord due to pernicious anaemia and cerebral and spinal tumours.
  - Spinal stenosis and active disease (e.g. spondylitis, tuberculosis, tumour) or recent trauma (e.g. fracture) in the vertebral column.
  - Septicaemia.
  - Pyogenic infection of the skin at or adjacent to the site of lumbar puncture.
  - Cardiogenic or hypovolaemic shock.

- Coagulation disorders or ongoing anticoagulation treatment.

**WARNINGS AND PRECAUTIONS :**

Intrathecal anaesthesia should only be undertaken by clinicians with the necessary knowledge and experience.  
Regional anaesthetic procedures should always be performed in a properly equipped and staffed area. Resuscitative equipment and drugs should be immediately available and the anaesthetist should remain in constant attendance.  
Intravenous access, e.g. an i.v. infusion, should be in place before starting the intrathecal anaesthesia. The clinician responsible should take the necessary precautions to avoid intravascular injection and be appropriately trained and familiar with the diagnosis and treatment of side effects, systemic toxicity and other complications. If signs of acute systemic toxicity or total spinal block appear, injection of the local anaesthetic should be stopped immediately.  
Like all local anaesthetic drugs, bupivacaine may cause acute toxicity effects on the central nervous and cardiovascular systems, if utilised for local anaesthetic procedures resulting in high blood concentrations of the drug. This is especially the case after unintentional intravascular administration or injection into highly vascular areas.  
Ventricular arrhythmia, ventricular fibrillation, sudden cardiovascular collapse and death have been reported in connection with high systemic concentrations of bupivacaine. Should cardiac arrest occur, a successful outcome may require prolonged resuscitative efforts. High systemic concentrations are not expected with doses normally used for intrathecal anaesthesia.  
There is an increased risk of high or total spinal blockade, resulting in cardiovascular and respiratory depression, in the elderly and in patients in the late stages of pregnancy. The dose should therefore be reduced in these patients.  
Intrathecal anaesthesia can cause hypotension and bradycardia. The risk of such effects can be reduced, e.g., by injecting a vasopressor. If hypotension develops it should be treated promptly with a sympathomimetic intravenously, repeated as necessary. Severe hypotension may result from hypovolaemia due to haemorrhage or dehydration, or aorto-caval occlusion in patients with massive ascites, large abdominal tumours or late pregnancy. Marked hypotension should be avoided in patients with hypovolaemia due to any cause can develop sudden and severe hypotension during intrathecal anaesthesia.  
Intrathecal anaesthesia can cause intercostal paralysis and patients with pleural effusions may suffer respiratory embarrassment. Septicaemia can increase the risk of intraspinal abscess formation in the postoperative period.  
Neurological injury is a rare consequence of intrathecal anaesthesia and may result in paraesthesia, anaesthesia, motor weakness and paralysis. Occasionally these are permanent.  
Before treatment is instituted, consideration should be taken if the benefits outweigh the possible risks for the patient.  
Patients in poor general condition due to ageing or other compromising factors such as partial or complete heart conduction block, advanced liver or renal dysfunction require special attention, although regional anaesthesia may be the optimal choice for surgery in these patients.

Patients treated with anti-arrhythmic drugs class III (e.g. amiodarone) should be kept under close surveillance and ECG monitoring considered, since cardiac effects may be additive.

**Interactions with Other Medicaments :**

Bupivacaine should be used with caution in patients receiving other local anaesthetics or agents structurally related to amide-type local anaesthetics, e.g. certain anti-arrhythmics, such as lidocaine and mexiletine, since the systemic toxic effects are additive.

Specific interaction studies with bupivacaine and anti-arrhythmic drugs class III (e.g. amiodarone) have not been performed, but caution is advised.

**PREGNANCY AND LACTATION :**

**Pregnancy**

There is no evidence of untoward effects in human pregnancy. In large doses, there is evidence of decreased pup survival in rats and an embryological effect in rabbits if bupivacaine is administered in pregnancy. Bupivacaine should not therefore be given in early pregnancy unless the benefits are considered to outweigh the risks.

It should be noted that the dose should be reduced in patients in the late stages of pregnancy.

**Breast-feeding**

Bupivacaine enters the mother's milk, but in such small quantities that there is generally no risk of affecting the child at therapeutic dose levels.

**EFFECTS ON ABILITY TO DRIVE AND USE MACHINES :**

Bupispine Heavy has minor influence on the ability to drive and use machines. Besides the direct anaesthetic effect, local anaesthetics may have a very mild effect on mental function and co-ordination even in the absence of overt CNS toxicity and may temporarily impair locomotion and alertness.

**SIDE EFFECTS :**

**General**

**Tabulated list of adverse reactions**

The adverse reaction profile for Bupivacaine Heavy is similar to those for other long acting local anaesthetics used for intrathecal anaesthesia.

Frequencies are defined as very common, common, uncommon, rare, very rare or not known (cannot be estimated from the available data).

**Table of Adverse Drug Reactions**

| System Organ Class       | Frequency Classification | Adverse Drug Reaction                                                              |
|--------------------------|--------------------------|------------------------------------------------------------------------------------|
| Immune system disorders  | Rare                     | Allergic reactions, anaphylactic shock                                             |
| Nervous system disorders | Common                   | Postdural puncture headache                                                        |
|                          | Uncommon                 | Paraesthesia, paresis, dysaesthesia                                                |
|                          | Rare                     | Total unintentional spinal block, paraplegia, paralysis, neuropathy, arachnoiditis |

|                                                 |             |                                         |
|-------------------------------------------------|-------------|-----------------------------------------|
| Cardiac disorders                               | Very Common | Hypotension, bradycardia                |
|                                                 | Rare        | Cardiac arrest                          |
| Respiratory, thoracic and mediastinal disorders | Rare        | Respiratory depression                  |
| Gastrointestinal disorders                      | Very Common | Nausea                                  |
|                                                 | Common      | Vomiting                                |
| Musculoskeletal and connective tissue disorders | Uncommon    | Muscle weakness, back pain              |
| Renal and urinary disorders                     | Common      | Urinary retention, urinary incontinence |

Adverse reactions caused by the drug *per se* are difficult to distinguish from the physiological effects of the nerve block (e.g. decrease in blood pressure, bradycardia, temporary urinary retention), events caused directly (e.g. spinal haematoma) or indirectly (e.g. meningitis, epidural abscess) by needle puncture or events associated to cerebrospinal leakage (e.g. postdural puncture headache).

**OVERDOSAGE :**

**Acute systemic toxicity**

BUPISPINE HEAVY, used as recommended, is not likely to cause blood levels high enough to cause systemic toxicity. However, if other local anaesthetics are concomitantly administered, toxic effects are additive and may cause systemic toxic reactions.

Systemic toxicity is rarely associated with spinal anaesthesia but might occur after accidental intravascular injection. Systemic adverse reactions are characterised by numbness of the tongue, light-headedness, dizziness and tremors, followed by convulsions and cardiovascular disorders.

**Treatment of acute systemic toxicity**

No treatment is required for milder symptoms of systemic toxicity but if convulsions occur then it is important to ensure adequate oxygenation and to arrest the convulsions if they last more than 15 - 30 seconds. Oxygen should be given by face mask and the respiration assisted or controlled if necessary. Convulsions can be arrested by injection of thiopental 100 - 150 mg intravenously or with diazepam 5 - 10 mg intravenously. Alternatively, succinylcholine 50 - 100 mg intravenously may be given but only if the clinician has the ability to perform endotracheal intubation and to manage a totally paralysed patient.

High or total spinal blockade causing respiratory paralysis should be treated by ensuring and maintaining a patent airway and giving oxygen by assisted or controlled ventilation. Hypotension should be treated by the use of vasopressors, e.g. ephedrine 10 - 15 mg intravenously and repeated until the desired level of arterial pressure is reached. Intravenous fluids, both electrolytes and colloids, given rapidly can also reverse hypotension.

**Paediatric population**

Adverse drug reactions in children are similar to those in adults, however, in children, early signs of local anaesthetic toxicity may be difficult to detect in cases where the block is given during sedation or general anaesthesia.

**INSTRUCTION FOR USE :**

BUPISPINE HEAVY should be inspected visually for discoloration and particulate matter prior to administration; solutions which are discolored or which contain particulate matter should not be administered. The solution should be used immediately after opening of the ampoule. Any remaining solution should be discarded. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

**EXCIPIENTS:**

Dextrose Monohydrate, Sodium hydroxide (pellets) and Water for Injection.

**STORAGE :**

Store below 30°C, protected from light.

**SHELF LIFE :**

24 Months

**PRESENTATION :**

Available in 5ml ampoule.

**REVISION DATE :** 16/06/2021

**PRODUCT REGISTRATION HOLDER :**

**Unimed Sdn. Bhd.**

No. 53, Jalan Tembaga SD 5/2B, Bandar Sri Damansara, 52200 Kuala Lumpur, Malaysia

Manufactured by :



**Troikaa Pharmaceuticals Ltd.**

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